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Three-dimensional multimodal imaging for predicting early recurrence of hepatocellular carcinoma after surgical resection



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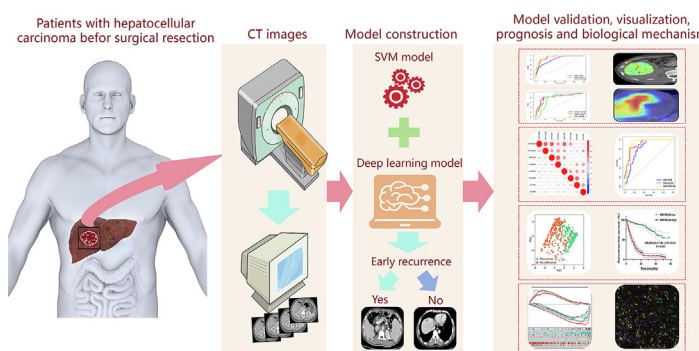
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HIGHLIGHTS

- A multimodal model (MM-RDLM) was developed to predict early recurrence of hepatocellular carcinoma after surgical resection.
- The MM-RDLM model achieved high predictive accuracy, with areas under the curve of 0.955 and 0.930 in the training and validation cohorts.
- Gene set enrichment analysis revealed enrichment of natural killer cell-mediated cytotoxicity pathway in MM-RDLM identified low-risk group.
- Multiplex immunohistochemistry showed higher CD3⁺, CD56⁺, and CD8⁺ cell densities in low-risk group, correlating with improved prognosis.
- Three-dimensional imaging can guide preoperative and postoperative medical decisions for patients with hepatocellular carcinoma.

GRAPHICAL ABSTRACT



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ABSTRACT

Background: High tumor recurrence after surgery remains a significant challenge in managing hepatocellular carcinoma (HCC). We aimed to construct a multimodal model to forecast the early recurrence of HCC after surgical resection and explore the associated biological mechanisms.

Materials and methods: Overall, 519 patients with HCC were included from three medical centers. 433

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patients from Nanfang Hospital were used as the training cohort, and 86 patients from the other two hospitals comprised validation cohort. Radiomics and deep learning (DL) models were developed using contrast-enhanced computed tomography images. Radiomics feature visualization and gradient-weighted class activation mapping were applied to improve interpretability. A multimodal model (MM-RDLM) was constructed by integrating radiomics and DL models. Associations between MM-RDLM and recurrence-free survival (RFS) and overall survival were analyzed. Gene set enrichment analysis (GSEA) and multiplex immunohistochemistry (mIHC) were used to investigate the biological mechanisms.

Results: Models based on hepatic arterial phase images exhibited the best predictive performance, with radiomics and DL models achieving areas under the curve (AUCs) of 0.770 (95 % confidence interval [CI]: 0.725–0.815) and 0.846 (95 % CI: 0.807–0.886), respectively, in the training cohort. MM-RDLM achieved an AUC of 0.955 (95 % CI: 0.937–0.972) in the training cohort and 0.930 (95 % CI: 0.876–0.984) in the validation cohort. MM-RDLM (high vs. low) was notably linked to RFS in the training (hazard ratio [HR] = 7.80 [5.74 – 10.61], $P < 0.001$) and validation (HR = 10.46 [4.96 – 22.68], $P < 0.001$) cohorts. GSEA revealed enrichment of the natural killer cell-mediated cytotoxicity pathway in the MM-RDLM low cohort. mIHC showed significantly higher percentages of CD3-, CD56-, and CD8-positive cells in the MM-RDLM low group.

Conclusion: The MM-RDLM model demonstrated strong predictive performance for early postoperative recurrence of HCC. These findings contribute to identifying patients at high risk for early recurrence and provide insights into the potential underlying biological mechanisms.

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Introduction

Hepatocellular carcinoma (HCC) is a highly fatal tumor. In 2022, statistics showed it ranked sixth among diagnosed malignancies globally and emerged as the third most significant contributor to cancer deaths [1]. Surgical resection is a common treatment for early-stage HCC [2,3]. Nevertheless, high rates of tumor recurrence after surgery present a significant challenge in managing HCC [4]. The recurrence rate within 5 years after surgery is approximately 70 % [5], with early recurrence (within 2 years) accounting for >70 % of cases [6–8]. Thus, recognizing patient groups with elevated risk for early recurrence remains essential for optimizing postoperative adjuvant treatments and improving survival outcomes.

Anthony et al. proposed the early recurrence after surgery for liver tumor (ERASL) models to assess the risk of early postoperative recurrence of liver cancer based on clinical variables [9]. However, the performance of the ERASL models remains suboptimal, with the highest C-index reported as 0.735. In addition, radiomics features, which could provide valuable insights into tumor characteristics [10], were not incorporated into these models. Unlike conventional imaging evaluations, radiomics analyzes the three-dimensional (3D) structure of the entire tumor [11,12] and converts digitally encoded medical images into quantitative features reflecting tumor pathophysiology [10,13]. Although some studies have explored radiomics for predicting postoperative recurrence of HCC, these efforts have primarily utilized two-dimensional arterial phase images [14]. Further investigation is required to assess the predictive potential of radiomics models utilizing 3D multiphase imaging data for early recurrence of HCC following surgery.

Deep learning (DL) identifies complex patterns in extensive datasets through a backpropagation algorithm and has substantially advanced the state-of-the-art in speech recognition, visual object recognition, object detection, and various other domains [15]. DL has been widely applied in tumor detection [16–18] and prognosis assessment [19–21], demonstrating satisfactory performance. Combining radiomics with DL holds great potential for predicting early postoperative recurrence of HCC.

In this study, we aimed to combine radiomics and DL to develop and validate models for assessing the risk of early recurrence of HCC following surgical resection using 3D-multimodal imaging.

Moreover, the underlying biological mechanisms associated with recurrence were explored.

Materials and methods

Study design

We aimed to construct and validate a multimodal model to forecast the early recurrence of HCC after surgical resection. An overview of the experimental design is illustrated in Fig. 1A. Four-phase contrast-enhanced computed tomography (CECT) images from patients in the training cohort were utilized to develop radiomics and DL models. The predictive performance of the models was evaluated, and radiomics features and key regions identified by the DL model were visualized. A multimodal model was also constructed by combining four radiomics models and four DL models. The predictive performance of all models was validated by employing the validation cohort. Furthermore, prognosis and underlying biological mechanisms were analyzed. The work has been reported in line with the REMARK criteria [22].

Study patients

The flowchart of patient recruitment is depicted in Fig. S1. Patients were retrospectively recruited based on the following criteria: pathologically confirmed HCC; stage I–III disease per the tumor, node, and metastasis (TNM) staging criteria (eighth edition); availability of hepatic arterial phase (HAP), portal venous phase (PVP), delayed phase (DP), and plain scan (PS) abdominal computed tomography (CT) images (including those obtained within 2 weeks prior to treatment); and receipt of standard radical resection. Patients were excluded if their Karnofsky Performance Status score was below 70, if they had undergone surgery or other therapies such as interventional therapy, immunotherapy, or oral targeted agents prior to the first surgical resection, or if their CT images were incomplete, of poor quality, or follow-up data were missing. Ultimately, 433 patients from Nanfang Hospital and 86 patients from Jiangxi Cancer Hospital and the First Affiliated Hospital, Hengyang Medical School, were included. In addition, 46 patients who underwent surgical treatment were identified from the Cancer Genome Atlas (TCGA).

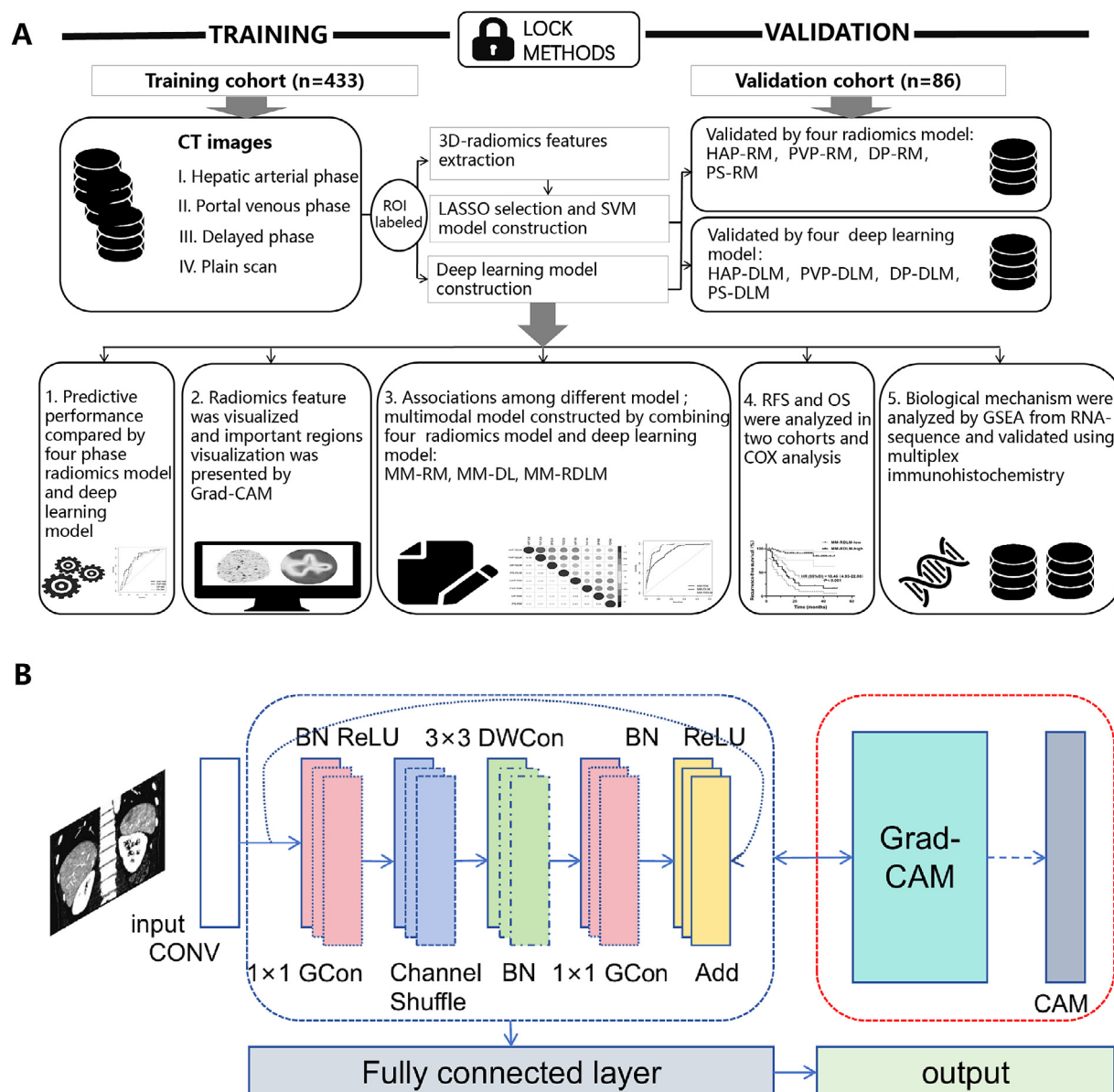


Fig. 1. Overview of the workflow in our study. (A) Study design of the multimodal model to estimate early recurrence. (B) Shuffle model construction. BN, batch normalization; ReLU, rectified linear unit; 1×1 GCon, 1×1 group convolution; 3×3 DWCon, 3×3 depthwise convolution; Grad-CAM, gradient-weighted class activation mapping; CAM, class activation map.

Clinical data collection, follow-up, and definitions

Clinical data recorded from the medical records of participating institutions included age, sex, alpha-fetoprotein (AFP) level (ng/mL), Child-Pugh classification, TNM stage, and early recurrence status.

Following surgical resection, patients were monitored for HCC recurrence through serum AFP levels, liver function tests, and CECT or magnetic resonance imaging executed every 6–12 weeks. The time span between the surgical date and initial detection of recurrence, metastasis, or final follow-up was designated as Recurrence-free survival (RFS). Tumor recurrence detected within 2 years post-surgery was classified as early recurrence. Overall survival (OS) was measured as the duration from the initiation of immunotherapy to either death or loss of follow-up.

Construction of radiomics models

CECT examinations were executed in clinical environments utilizing multi-slice spiral CT scanners. Two experienced radiologists,

blinded to diagnosis, segmented tumors manually using ITK-SNAP software to delineate the region of interest (ROI) on each slice of the liver CECT images for each patient. The two radiologists repeated the segmentation in 50 randomly selected cases after 1 month to evaluate intra- and inter-observer reproducibility. Inter- and intra-class correlation coefficients (ICCs) greater than 0.75 are considered to indicate good consistency. A senior radiologist then reviewed the segmentations and made any necessary adjustments.

To address variations in radiological properties resulting from differences in reconstruction slice thickness and pixel size, all CT images in this study underwent reconstruction with a voxel size of $1 \times 1 \times 1$ mm³. Furthermore, the ROI was standardized to 64 Gy levels to account for disparities among CT scanners. Radiomics features of the ROI were computed utilizing Python (version 3.9.1) and the “Pyradiomics” package. Extracted features included first-order statistics, shape features, and texture features derived from the gray level co-occurrence matrix (GLCM), gray level size zone matrix (GLSZM), gray level run length matrix (GLRLM), neigh-

boring gray tone difference matrix, and gray level dependence matrix.

Radiomics features with high correlations (correlation coefficient > 0.75) were excluded. Least absolute shrinkage and selection operator (LASSO) regression and 5-fold cross-validation were then utilized to determine features significantly associated with early postoperative recurrence (Fig. S2). We used Lambda.1se as the λ value and selected the optimal features. Using the selected features, the support vector machine (SVM) algorithm was implemented to compute the radiomics score. Based on 5-fold cross validation, we used GridSearchCV for parameter optimization to evaluate the parameter combination. The SVM specifications were configured as follows: SVM-Type: eps-regression; SVM-Kernel: radial; Cost: 1; Gamma: 0.125; Epsilon: 0.1; Number of Support Vectors: 158.

To increase the interpretability of the radiomics features, we performed a visual analysis on them. The visualization of radiomics features needs to combine the spatial information of CT images and the radiomics features (such as GLCM, GLRLM and GLSZM), and map the radiomics features into 3D space. Firstly, voxel-level feature mapping, the calculation results of texture features (such as entropy, energy) are mapped back to the corresponding voxel positions of the original 3D images, and the feature volume data is generated. Secondly, pseudo-color overlay, using color mapping (such as Jet and Viridis) to convert feature values into colors, overlays on grayscale images, and adjusts the transparency to retain the anatomical structure. Thus, the pseudo-color image is overlaid on the CT image to form a visualization of the radiomics features, with blue indicating low radiomics feature values and red indicating high radiomics feature values.

Construction of DL models

The CT scans were annotated utilizing MATLAB 2018a (<https://www.mathworks.cn/>) and adjusted to dimensions appropriate for DL model input. ShuffleNet, a deep convolutional neural network architecture focused on feature extraction from bounding box images, was selected as the underlying framework (Fig. 1B). ShuffleNet includes key modules such as pointwise group convolution, channel shuffle, and depthwise convolution, all of which play critical roles in the feature extraction process.

To optimize the model's effectiveness, a transfer learning approach was implemented by initializing the ShuffleNet model with weights pretrained on an HCC dataset. This strategy leverages knowledge from a large and diverse dataset, significantly accelerating convergence and improving accuracy for the specific task. In addition, a fully connected network with an extra hidden layer comprising 1,024 nodes was introduced to further integrate the extracted features. This hidden layer serves an essential function in combining features from the convolutional network. The Softmax activation mechanism was applied to transform the output into a probability distribution, enabling ultimate categorization into two categories. This strategic design enhances the performance of the model, guaranteeing precise classification outcomes. The model's training was configured with a maximum of 200 epochs, providing sufficient iterations for effective feature learning and thorough convergence.

To enhance model interpretability, gradient-weighted class activation mapping (Grad-CAM) was applied to examine significant areas within the CECT images. Grad-CAM works by quantifying the correlation between category predictions and feature maps from convolutional layers, producing class activation maps. This technique highlights critical features and regions involved in the decision-making process of DL models, offering a powerful tool for understanding and interpreting model behavior.

Development of multimodal models

Correlation heat maps were used to describe the associations among different models. Radiomics and DL models were combined to develop multimodal radiomics (MM-RM) and multimodal DL (MM-DLM) models, respectively. Furthermore, all individual models were integrated to construct a comprehensive multimodal model based on radiomics and DL models (MM-RDLM). The MM-RDLM model was designed to generate early recurrence probability scores for patients with HCC. Based on a probability score cutoff value of 0.22, all patients were split into MM-RDLM low and MM-RDLM high cohorts.

Gene set enrichment analysis (GSEA)

The association between signaling pathways, gene expression, and the MM-RDLM model was evaluated using GSEA. CT images and RNA-sequence data were collected from 46 individuals with HCC in TCGA, comprising 27 individuals in the MM-RDLM high cohort and 19 in the MM-RDLM low cohort. Differential gene expression analysis between the two cohorts was performed using the “edge” software, with criteria of a fold change ≥ 1.5 and $P < 0.05$. The biological pathways linked to the observations were identified through the MSigDB gene set database (<https://ngdc.cncb.ac.cn/databasecommons/database/id/1077>).

Multiplex immunohistochemistry (mIHC)

mIHC was performed to evaluate the spatial and quantitative expression of multiple protein targets within tumor tissue samples. Paraffin-embedded sections underwent deparaffinization and rehydration steps. Citrate buffer (pH 6.0) was utilized for antigen recovery, succeeded by suppression of endogenous peroxidase utilizing 3 % H₂O₂ over 15 min. After sealing with 3 % bovine serum albumin for 30 min, the slides underwent overnight exposure to primary antibodies: CD3 (ab16669, 1:500), CD8 (66868-1-2g, 1:1000), CD56 (ab313779, 1:500), and Foxp3 (ab215206, 1:200). Subsequently, the sections underwent washing with phosphate-buffered saline and were exposed to a horseradish peroxidase-conjugated secondary antibody for 50 min, followed by visualization using fluorophore-based tyramide signal amplification. After microwave treatment and serum blocking, panoramic scans of the slides were performed for analysis.

Statistical analysis

The study population features were summarized employing descriptive statistics. Variables were reported as n (%) and analyzed utilizing the chi-square test. A receiver operating characteristic (ROC) curve analysis was executed to examine the models' predictive efficiency. Scatter plots were used to visualize MM-RDLM model scores for recurrent and non-recurrent cohorts. Kaplan–Meier curves for RFS and OS were generated based on MM-RDLM cohort classifications. The clinical applicability of the multimodal models was assessed using decision curve analysis. Based on the Benjamini-Hochberg method, multiple comparison adjustments were made for GSEA and mIHC multigroup analyses in the fdrtool package. DeLong's test was used to compare the AUC values of different models.

The DL framework was implemented using PyTorch 2.0 (<https://pytorch.org/>) in combination with Python 3.9 (<https://www.python.org/>). This selection offered adaptability and code clarity while harnessing the robust capabilities of PyTorch, improving the efficiency and convenience of the experimental design and implementation. Python was also applied for all statistical analy-

ses. For all tests, a two-sided *P*-value < 0.05 was deemed statistically significant.

Results

Characteristics of the study cohorts

A total of 519 patients with HCC from three medical centers were included, with 433 patients from Nanfang Hospital forming the training cohort and 86 patients from the other two centers comprising the validation cohort. The characteristics of the two cohorts are depicted in Table 1.

In the training cohort, most patients were male (87.76 %) and aged <60 years (76.44 %). Liver function was classified as Child-Pugh A in 396 patients, and 335 patients had stage I tumors according to TNM classification. During follow-up, 206 patients (177 in the training cohort and 29 in the validation cohort) experienced early recurrence following curative surgical resection. The patient's characteristics in the training and validation cohorts did not differ significantly.

Performance of radiomics models

Liver CECT images from patients with HCC were utilized to extract radiomics features across four phases. In the training cohort, LASSO regression was applied to ascertain features significantly linked to early recurrence of HCC. Ultimately, 17 features were selected from the HAP, 13 from the PVP, 8 from the DP, and 7 from the PS. Details of these selected features are provided in Table S1.

Using the selected features, four radiomics models (HAP-RM, PVP-RM, DP-RM, and PS-RM) were constructed with the SVM algorithm. The predictive performance of these models was evaluated utilizing ROC curve analysis. In the training cohort, the HAP-RM model displayed the greatest predictive performance for early recurrence of HCC, with an area under the curve (AUC) of 0.770 (95 % confidence interval [CI]: 0.725–0.815; Fig. 2A). Similarly, HAP-RM demonstrated superior predictive ability in the validation cohort, achieving an AUC of 0.765 (95 % CI: 0.660–0.869; Fig. 2B). Detailed AUC values for all radiomics models are presented in Table S2.

To further illustrate model performance, radiomics features from preoperative CT images of four patients were visualized. Both patient 1 and patient 2 were in TNM stage IB with similar clinical characteristics. Patient 2, who experienced longer RFS, had significantly lower radiomics model scores (HAP-RM = −0.04, PVP-RM = 0.05, DP-RM = −0.02, PS-RM = 0.04) compared with patient 1 (HAP-RM = 0.94, PVP-RM = 0.67, DP-RM = 0.72, PS-RM = 0.97), as shown in Fig. 2C. Meanwhile, we presented the radiomics feature of two HCC patients (patient 3 and patient 4) with stage IIIA, which showed similar clinical data and significant survival differences (Fig. S3).

Performance of DL models

Four deep convolutional neural network models (HAP-DLM, PVP-DLM, DP-DLM, and PS-DLM) were developed using four-phase CECT images to accurately predict early postoperative recurrence. To improve the performance and stability of the models, ShuffleNet was chosen as the foundational framework, optimized for extracting features from bounding box images.

The ROC curve analysis suggested that the HAP-DLM model demonstrated superior predictive ability compared with other models, achieving an AUC of 0.846 (95 % CI: 0.807–0.886) in the training cohort (Fig. 3A). In the validation cohort, however, the

Table 1
Characteristics of patients in the training and validation cohorts.

Characteristic	Training cohort (n = 433)	Validation cohort (n = 86)	<i>P</i> value
Age (years)			0.316
≤60	331 (76.44 %)	70 (81.40 %)	
>60	102 (23.56 %)	16 (18.60 %)	
Sex			0.295
Female	53 (12.24 %)	14 (16.28 %)	
Male	380 (87.76 %)	72 (83.72 %)	
AFP level (ng/ml)			0.470
≤20	177 (40.88 %)	31 (36.05 %)	
>20	256 (59.12 %)	55 (63.95 %)	
Child-Pugh classification			0.999
A	396 (91.45 %)	79 (91.86 %)	
B	37 (8.55 %)	7 (8.14 %)	
TNM stage			0.707
I	335 (77.36 %)	67 (77.90 %)	
II	80 (18.47 %)	17 (19.76 %)	
III	18 (4.17 %)	2 (2.34 %)	
Early recurrence			0.215
Yes	177(40.88 %)	29 (33.72 %)	
No	256 (59.12 %)	57 (66.28 %)	

AFP, alpha-fetoprotein; TNM, tumor, node and metastasis.

PVP-DLM model exhibited the highest predictive capacity for early recurrence, with an AUC of 0.825 (95 % CI: 0.722–0.928), sensitivity of 82.76 % (95 % CI: 65.45–92.40), and specificity of 84.21 % (95 % CI: 72.64–91.46; Fig. 3B). Detailed performance metrics for all DL models were provided in Table S3.

Further, to enhance the interpretability of the models, important regions within the CECT images were visualized using Grad-CAM. The gradient size functioned as a measure for assessing the “importance” of individual nodes or voxels in relation to the ultimate prediction layer. In the Grad-CAM visualizations, red hues signified regions with elevated predictive probability scores, whereas highlights marked the significant areas that influenced the DL model in predicting early recurrence. Conversely, blue hues represented areas with lower predictive probability scores according to the DL model (Fig. 3C, Fig. S4).

Performance of multimodal models

The associations among different models were assessed using correlation heatmaps. In the training cohort, radiomics models demonstrated strong correlations with other radiomics models (all *P* < 0.001), and the results among deep learning models of different phases are similar (all *P* < 0.001) (Fig. 4A). Comparable associations were observed in the validation cohort (Fig. 4B). The strong correlations observed across different imaging phases suggest that these models achieve consistent performance in capturing reproducible radiomic signatures throughout various CT contrast phases. Based on these findings, we developed the MM-RM model by combining the four radiomics models and the MM-DLM model by combining the four DL models. The predictive performance of the multimodal models surpassed that of the individual models, with MM-RM achieving an AUC of 0.836 (95 % CI: 0.799–0.873) and MM-DLM achieving an AUC of 0.927 (95 % CI: 0.903–0.950) in the training cohort.

All radiomics and DL models were combined to develop the final multimodal model (MM-RDLM) to enhance predictive accuracy. The predictive performance of MM-RDLM surpassed that of the other models (Fig. 4C and D). In the training cohort,

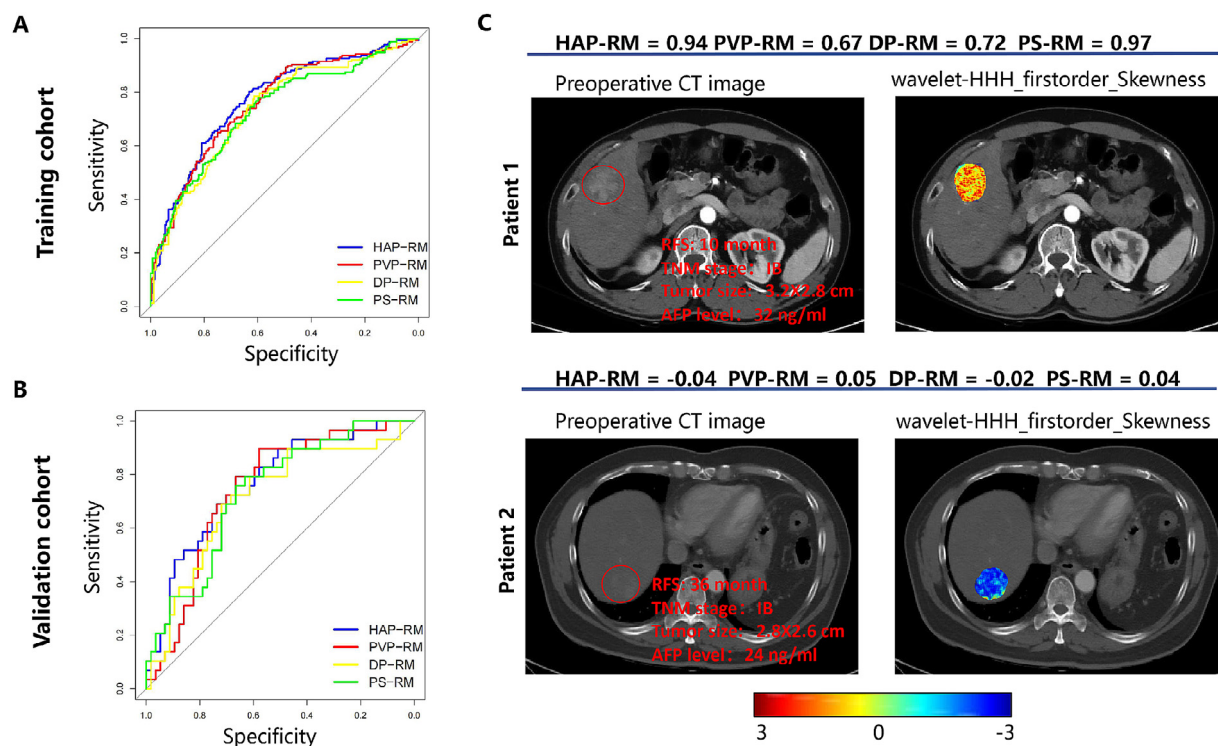


Fig. 2. Validation of radiomics models for early recurrence prediction. (A) ROC curves for early recurrence prediction using radiomics models in the training cohort. (B) ROC curves for early recurrence prediction employing radiomics models in the validation cohort. (C) Visualization of radiomics features in two patients. ROC, receiver operating characteristic; HAP-RM, hepatic arterial phase radiomics model; PVP-RM, portal venous phase radiomics model; DP-RM, delayed phase radiomics model; PS-RM, plain scan radiomics model; RFS, recurrence-free survival; CT, computed tomography; wavelet-HHH, wavelet high-high-high decomposition; firstorder_Skewness, first-order skewness.

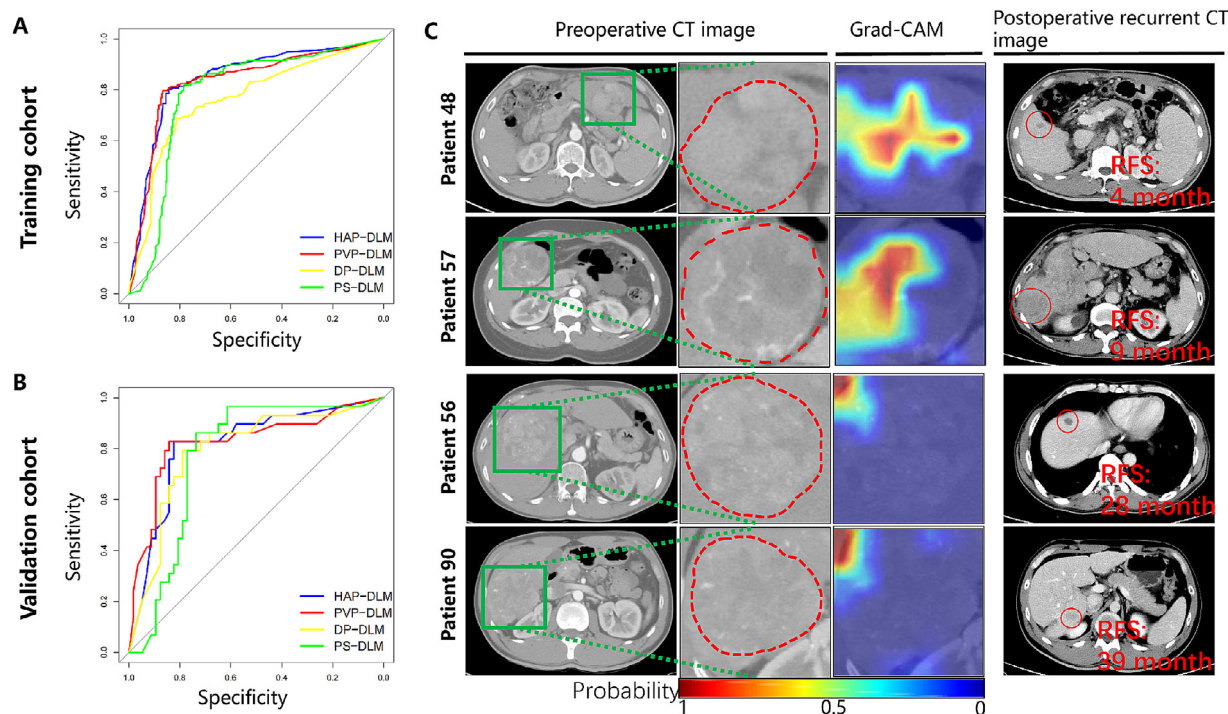


Fig. 3. Validation of DL models for early recurrence prediction. (A) ROC curves for early recurrence prediction using DL models in the training cohort. (B) ROC curves for early recurrence prediction using DL models in the validation cohort. (C) Visualization of important regions in DL models. Column 1 shows preoperative CT images of four patients. Column 2 displays Grad-CAM images highlighting darker red areas, which indicate regions contributing most to the outputs of the final prediction layer. Red denotes high predictive scores for early recurrence, whereas blue indicates low predictive scores. Column 3 shows postoperative recurrent CT images of four patients. DL, deep learning; ROC, receiver operating characteristic; CT, computed tomography; Grad-CAM, gradient-weighted class activation mapping; HAP-DLM, hepatic arterial phase deep learning model; PVP-DLM, portal venous phase deep learning model; DP-DLM, delayed phase deep learning model; PS-DLM, plain scan deep learning model; RFS, recurrence-free survival. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

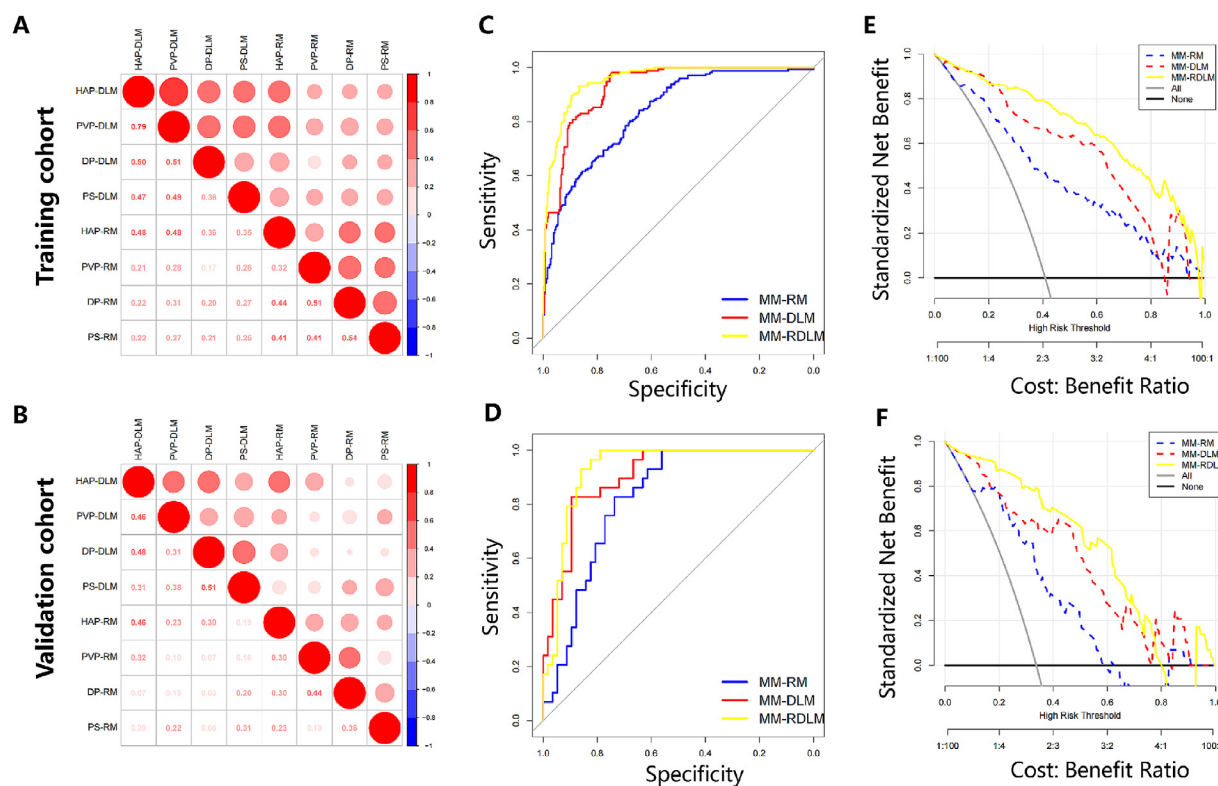


Fig. 4. Validation of multimodal models for early recurrence prediction. (A, B) Correlation heatmaps between radiomics and DL models in the training and validation cohorts. (C, D) ROC curves for early recurrence prediction using multimodal models in the training and validation cohorts. (E, F) DCA for early recurrence prediction using multimodal models in the training and validation cohorts. DL, deep learning; ROC, receiver operating characteristic; DCA, decision curve analysis; HAP-DLM, hepatic arterial phase deep learning model; PVP-DLM, portal venous phase deep learning model; DP-DLM, delayed phase deep learning model; PS-DLM, plain scan deep learning model; HAP-RM, hepatic arterial phase radiomics model; PVP-RM, portal venous phase radiomics model; DP-RM, delayed phase radiomics model; PS-RM, plain scan radiomics model; MM-RM, multimodal radiomics model; MM-DLM, multimodal deep learning model; MM-RDLM, multimodal model combining radiomics and deep learning models.

MM-RDLM achieved an AUC of 0.955 (95 % CI: 0.937–0.972), whereas in the validation cohort, it achieved an AUC of 0.930 (95 % CI: 0.876–0.984; Table S4). Moreover, decision curve analysis demonstrated that MM-RDLM provided a greater net benefit across a threshold probability range of approximately 0.10–0.80 for all cohorts (Fig. 4E and F). Additionally, we integrated clinical features significantly associated with early recurrence to develop a comprehensive model. In the training cohort, based on the DeLong test, we did not observe a significant improvement in the predictive performance of the comprehensive model (AFP + TNM + MMRDLM) compared to the MM-RDLM model ($P = 0.748$), and the result was similar in the validation cohort ($P = 0.821$) (Fig. S5).

Prognostic evaluation of MM-RDLM

MM-RDLM effectively distinguished between postoperative recurrent and non-recurrent patients, as visualized using scatter plots (Fig. 5A and B). The model generated a recurrence risk score for each patient, utilizing the maximum Youden index to establish the optimal threshold value. Based on this threshold, individuals were split into high and low MM-RDLM cohorts. Univariate and multivariate Cox regression analyses identified the dichotomized MM-RDLM risk score and AFP level as independent predictors of RFS and OS in patients with HCC (Tables S5 and S6).

To examine the link between the MM-RDLM model and clinical outcomes, we evaluated RFS and OS in both cohorts. In the training cohort, individuals in the MM-RDLM low cohort exhibited markedly extended RFS compared with those in the MM-RDLM high cohort (hazard ratio [HR]: 7.80, 95 % CI: 5.74–10.61, $P < 0.001$; Fig. 5C). Similarly, OS was notably longer in the MM-RDLM low

cohort versus the MM-RDLM high cohort (HR: 3.88, 95 % CI: 2.83–5.32, $P < 0.001$; Fig. 5E). In the validation cohort, patients in the MM-RDLM low cohort also demonstrated a more favorable prognosis (Fig. 5D and F). Furthermore, data from 46 patients with HCC from the TCGA database confirmed that MM-RDLM accurately predicted the prognosis of these patients (Fig. S6).

Biological mechanism analyses

To investigate the biological mechanisms underlying the ability of MM-RDLM to predict early postoperative recurrence of HCC, GSEA was performed to identify pathway enrichment differences between the MM-RDLM low and MM-RDLM high cohorts using RNA-sequence data. The analysis suggested that the natural killer (NK) cell-mediated cytotoxicity pathway was significantly enriched in patients with HCC in the MM-RDLM low cohort (Fig. 6A).

The expression of CD3, CD56, and CD8 was further evaluated between the MM-RDLM low and MM-RDLM high cohorts using mIHC. In the MM-RDLM low cohort, the expression of CD3, CD56, and CD8 was significantly increased, with a higher density of positive cells compared with the MM-RDLM high cohort (Fig. 6B). Quantitative analysis confirmed that the percentage of CD3-, CD56-, and CD8-positive cells was notably elevated in the MM-RDLM low cohort versus the MM-RDLM high cohort (Fig. 6C).

Discussion

This multicenter study developed and validated multimodal models integrating radiomics and DL to predict early postoperative

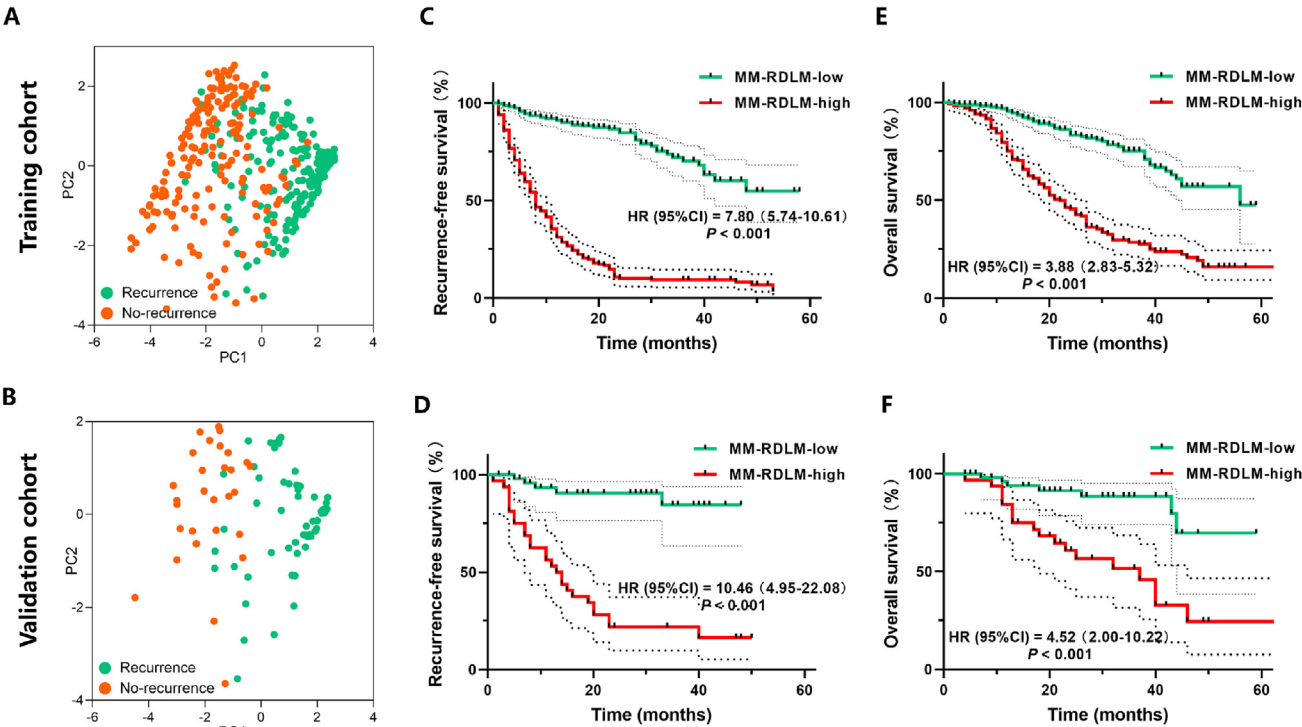


Fig. 5. Prognostic evaluation using the MM-RDLM model. (A, B) Scatter plots of MM-RDLM model scores for recurrence and non-recurrence cohorts in the training and validation cohorts. (C, D) Prediction of RFS using MM-RDLM in the training and validation cohorts. (E, F) Prediction of OS using MM-RDLM in the training and validation cohorts. MM-RDLM, multimodal model combining radiomics and deep learning models; RFS, recurrence-free survival; OS, overall survival; PC1, principal component 1; PC2, principal component 2; HR, hazard ratio; CI, confidence interval.

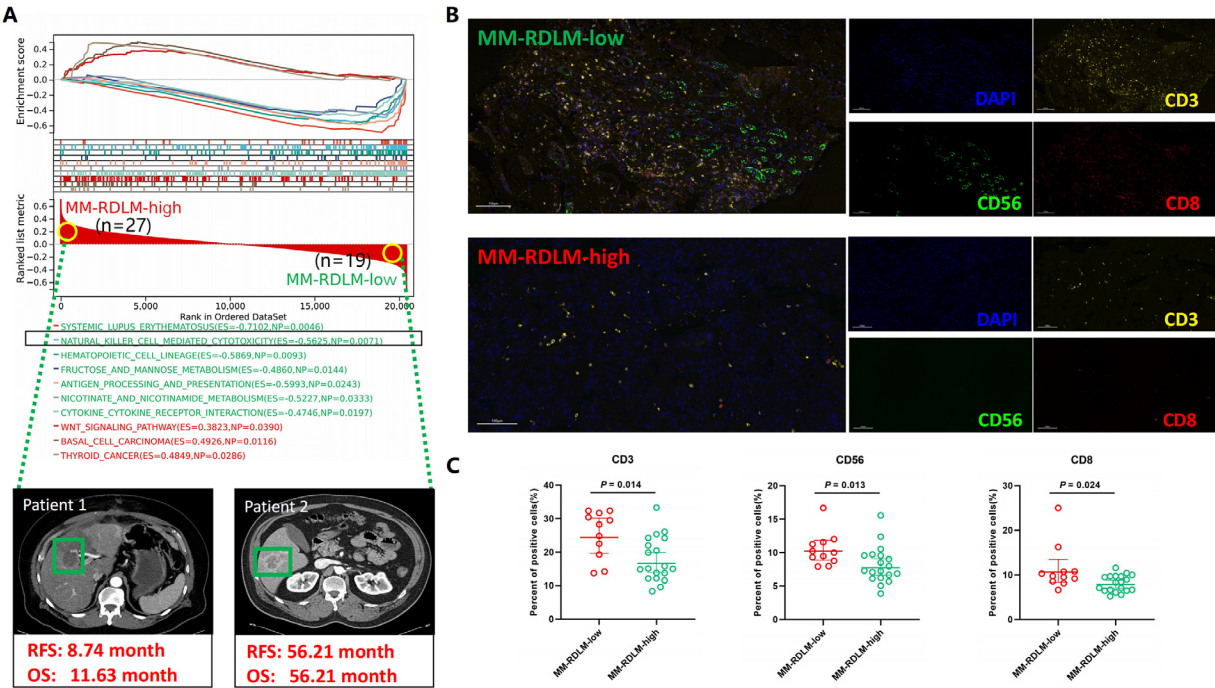


Fig. 6. Biological mechanisms associated with the MM-RDLM model. (A) GSEA results comparing the MM-RDLM high and low cohorts. (B) mIHC images of DAPI, CD3, CD8, and CD56 staining in HCC tissues from the MM-RDLM high and low cohorts. Scale bar: 100 μ m. (C) Percentage of positive cells in tumor tissues assessed in the MM-RDLM high and low cohorts. MM-RDLM, multimodal model combining radiomics and deep learning models; GSEA, gene set enrichment analysis; mIHC, multiplex immunohistochemistry; DAPI, 4',6-diamidino-2-phenylindole; HCC, hepatocellular carcinoma; RFS, recurrence-free survival; OS, overall survival.

recurrence of HCC. The MM-RDLM model demonstrated strong performance, enabling clinicians to establish well-informed surgical decisions regarding the optimal treatment plan and timing. In

addition, this model may guide individualized treatment strategies and facilitate survival monitoring for individuals with operable HCC.

The early recurrence rate of HCC remains high after surgery resection, with approximately 40 % of patients in this study experiencing early recurrence. CECT is a standard diagnostic tool for patients with HCC, and its imaging potential is increasingly recognized [23,24]. Using four-phase 3D CECT data, we constructed radiomics models with SVM and DL models with ShuffleNet, combining them into the MM-RDLM model. This model displayed excellent predictive capability, showing AUCs of 0.955 and 0.930 for the training and validation cohorts. In comparison, the genetic algorithm for predicting recurrence after surgery of liver cancer (GARSL) model, developed using both clinical and radiomics features, achieved an AUC of only 0.739 [7]. Furthermore, GARSL and many previously reported models were based on single-center data, limiting their generalizability and clinical applicability [25–27]. A preoperative radiomics model developed by Ji et al. using multicenter data achieved respective AUCs of 0.82 and 0.84 in the development and test cohorts [28], both lower than those achieved by MM-RDLM. Overall, MM-RDLM demonstrated strong predictive performance and robustness, highlighting the value of combining radiomics and DL features for predicting early postoperative recurrence of HCC.

Radiomics extracts high-throughput quantitative features from medical images [29,30], with complexity and high dimensionality surpassing human cognitive capabilities [24,31]. We visualized selected radiomics features to improve understanding. Conversely, DL models face challenges related to the “black box” effect, as their low interpretability and opacity remain significant issues [32,33]. Interpretation is particularly challenging when DL models are applied in the field of medicine [34]. Grad-CAM was used to enhance interpretability by identifying important areas analyzed by MM-RDLM, showing that the DL model primarily focused on tumor regions when predicting recurrence, providing insights into its decision-making process.

To further enhance the interpretability of the MM-RDLM model, we analyzed potential biological mechanisms. GSEA revealed that the NK cell-mediated cytotoxicity pathway showed enrichment within the MM-RDLM low cohort. Patients in this cohort also exhibited longer RFS and OS. NK cells serve an essential function in controlling HCC, and the density of tumor-infiltrating NK cells is positively associated with the OS rate of individuals with HCC [35,36]. However, NK cells are often dysfunctional in the HCC microenvironment owing to factors such as decreased abundance and impaired tumor recognition [37]. Enhancing NK cell function within the HCC microenvironment may therefore help to reduce postoperative recurrence and prolong survival. A clinical trial in Korea (NCT02008929) investigated allogeneic NK cell transfer as a strategy to combat HCC recurrence after curative resection. The findings indicated that immunotherapy using *ex vivo*-expanded allogeneic NK cells in post-hepatectomy patients was safe and provided preliminary evidence of efficacy in reducing recurrence among patients with HCC following curative liver resection [38]. Further prospective clinical studies are needed to evaluate whether NK cell immunotherapy can offer significant clinical benefits for post-hepatectomy patients.

The mIHC results showed the substantial infiltration of CD3-, CD56-, and CD8-positive cells in tumor tissues from the MM-RDLM low cohort. The CD3 protein complex plays a critical role in the maturation of immature precursor thymocytes, with CD3 δ exhibiting fundamental importance for the growth and differentiation of human T lymphocytes [39]. In addition, CD3 molecules transduce activation signals from T-cell receptor recognition of antigens to T cells [40]. CD8-positive T lymphocytes are key players in killing cancer cells by releasing cytotoxic molecules and cytokines [41,42]. Previous studies have observed a significant increase in tumor volume and weight following the depletion of CD8⁺ T cells. Certain specialized diets, such as serine/glycine-restricted

(–SG) diets [43], can promote the expression of CD8⁺ T cells, thereby enhancing antitumor immunity. In the future, these dietary interventions may also serve as supportive treatments to prevent postoperative recurrence. It has been confirmed that a high density of CD3- or CD8-positive T cells is linked to better survival outcomes [44,45]. In individuals with HCC undergoing surgery, a high density of CD3- and CD8-positive cells strongly correlates with prolonged RFS [46]. Similarly, Liang et al. observed that patients with a high density of CD3- and CD8-positive cells experienced longer RFS and OS in colorectal cancer with hepatic metastasis [47]. Regarding CD56, reduced levels of CD56-positive T cells and NK cells are associated with increased susceptibility to liver cancer in patients with cirrhosis [48]. Research on solid tumors has also revealed a significant correlation between CD56 and better OS and progression-free survival [49]. Our findings align with these observations, as patients with higher percentages of CD3-, CD56-, and CD8-positive cells had longer RFS and OS. These findings offer a crucial understanding of the possible biological pathways that drive the multimodal model.

This investigation had certain constraints. Given the retrospective nature of this study, the limitations of data collection are inherent and there may be selection bias. Despite it being a multicenter study, the study population remained relatively limited, especially the validation cohort. Subsequent investigations incorporating more extensive patient cohorts are essential to validate the efficacy of the predictive models. Moreover, the manual segmentation of HCC tumors was time-consuming; however, advances in automated segmentation methods are expected to significantly reduce this burden in the future. Finally, all patients included in the study were from China, with most likely having a hepatitis B background. This may limit the generalizability of the findings. Further validation is required in patients with different backgrounds, such as those with fatty liver disease.

Conclusion

In this study, we combined radiomics and DL methods to establish a multimodal model for predicting early postoperative recurrence in individuals with HCC. The MM-RDLM model demonstrated strong performance in stratifying risk for RFS and OS, effectively identifying patients at high risk of early recurrence. This study offers a novel approach to predicting postoperative recurrence in HCC and provides valuable insights into the underlying biological mechanisms. These findings have significant implications for the postoperative management and clinical decision-making process for patients with HCC undergoing surgery.

Compliance with ethics requirements

All procedures involving human participants were conducted in accordance with the ethical standards of the institutional and/or national research committee, as well as the 1975 Declaration of Helsinki and its later amendments or comparable ethical guidelines. The study design was approved by the Medical Ethics Committee of the Nanfang Hospital, Southern Medical University (approval number: NFEC-2021-048), Jiangxi Cancer Hospital (approval number: 2022ky087), and the First Affiliated Hospital, Hengyang Medical School, University of South China (approval number: ISL2024224). The need for obtaining informed consent was waived owing to the retrospective study design.

CRediT authorship contribution statement

Jie Peng: Conceptualization, Resources, Investigation, Methodology, Formal analysis, Visualization, Writing – review & editing,

Funding acquisition. **Jiaren Wang**: Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Hongbo Zhu**: Conceptualization, Resources, Writing – review & editing. **Pu Jiang**: Data curation, Writing – original draft. **Ji Xia**: Investigation, Methodology, Data curation. **Hao Cui**: Validation. **Chang Hong**: Validation, Funding acquisition. **Lin Zeng**: Validation. **Ruining Li**: Validation. **Yan Li**: Validation. **Shengxing Liang**: Validation. **Qijie Deng**: Validation. **Huangying Deng**: Validation. **Hengtian Xu**: Validation. **Hanzhi Dong**: Resources, Supervision. **Lushan Xiao**: Conceptualization, Writing – review & editing, Funding acquisition, Supervision. **Li Liu**: Resources, Funding acquisition, Supervision.

Data availability

The original CT scans and medical records utilized in this investigation cannot be openly shared owing to institutional policies and confidentiality concerns regarding patient information. The datasets supporting the conclusions drawn from this research may be obtained by contacting the corresponding author with appropriate justification.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jare.2025.06.031>.

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